From Differential Privacy to Concentrated Privacy

How the privacy cost of many releases adds up, derived from first principles

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When you publish many statistics about the same people, how much privacy do they lose in total? For a reader who knows basic probability and nothing about privacy, this note derives why many small privacy losses add up to much less than their worst case, and how concentrated and zero-concentrated differential privacy make that saving easy to compute.

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A statistic computed from a dataset, such as a count, reveals a little about each person in it. Differential privacy limits how much. It asks that the published output be almost as likely whether or not any one person's record is in the data. Random noise makes this possible. Without noise, adding one person to a count changes the answer by exactly one, and anyone who compares two releases can see it. With noise, the two possible answers overlap, so one output cannot reliably tell them apart.

This note asks how the privacy cost of many releases adds up. The simplest answer is that the costs add, so a hundred releases cost a hundred times as much as one. That answer is safe but pessimistic, because it charges every release its worst case. In fact the cost of a release is random, usually far below its worst case and sometimes negative. So the total is a sum of small average costs plus a fluctuation that grows only like the square root of the number of releases.

You need only basic probability: densities, expectations, variances, the normal distribution and Markov's inequality. Table 1 gives the question that each section answers, and Table 2 lists the symbols.

Table 1: The question each section answers.

Section	The question it answers
Section 1	What does a privacy guarantee compare, output by output?
Section 2	What is the strongest guarantee, and how much does it cost over many releases?
Section 3	What guarantee does Gaussian noise give, since its loss is unbounded?
Section 4	How do you prove such a guarantee?

Section	The question it answers
Section 5	Why do many small losses add up to much less than their worst case?
Section 6	Can the facts behind that saving become the definition of privacy?
Section 7	What definition makes the saving easy to compute?
Section 8	How do the four definitions compare?
Section 9	What do they give for a hundred Gaussian releases?

Table 2: The symbols the note uses. Each is also explained where it first appears.

Symbol	Meaning
D, D'; $D \sim D'$	two neighbouring datasets, which means that one contains a person's record and the other does not
M	a mechanism, the random procedure that computes the released output
f, Δf	the statistic to release, and its sensitivity: how much one record can change it
d	the difference $f(D) - f(D')$ between the two true answers
P, Q	the distributions of the released output when the data are D and D'
p, q	their densities
L	the privacy loss, $\ln(p/q)$ at the released output
ε, δ	the parameters of a differential-privacy guarantee
δ'	a small probability that you choose yourself when you derive one guarantee from another
E_{e^ε}, $\delta(\varepsilon)$	the hockey-stick divergence, and the smallest δ that works for a given ε, called the privacy profile
b, σ	the scale of Laplace noise, and the standard deviation of Gaussian noise
k	the number of releases
μ, τ	the two parameters of concentrated differential privacy, a mean and a spread
ξ, ρ	the two parameters of zero-concentrated differential privacy
λ, α	the argument of a moment generating function, and the order of a Rényi divergence

1 The privacy loss

Every definition in this note compares two probability distributions of the released output, one for each of two datasets. How do you measure, output by output, how much they differ? The answer is one number per output, the privacy loss.

1.1 Two output distributions

Take two datasets D and D' that are the same except that D contains one person's record and D' does not. Such datasets are **neighbouring**, written $D \sim D'$. Some authors instead call two datasets neighbouring when one record replaces another. The derivations hold for either choice, but the numbers in the examples depend on it, and this note always adds or removes a record.

A **mechanism** M is a random procedure that takes a dataset and releases an output. On D its output has a probability distribution P, and on D' another distribution Q. Both depend on the pair of datasets, although the notation does not show it. If they were identical, the output would reveal nothing about whether the person is in the data.

The running example is a counting query. The statistic $f(D)$ is the number of people in dataset D who have some medical condition, and adding or removing one person changes it by at most one. In general, the **sensitivity** Δf of a statistic f is the most it can change between neighbouring datasets:

$$\Delta f = \max_{D \sim D'} |f(D) - f(D')|. \tag{1}$$

So a count has $\Delta f = 1$. To release it, a mechanism adds random noise. With noise from the Laplace distribution of scale b, which has density $e^{-|z|/b}/(2b)$, the output on dataset D has the density p below:

$$p(y) = \frac{1}{2b} \exp\left(-\frac{|y - f(D)|}{b}\right). \tag{2}$$

This is a peaked curve centred at the true count $f(D)$. On D' the output density q is the same curve, centred at $f(D')$ instead.

1.2 The privacy loss

To see how much one output reveals, imagine an observer who knows everything about the data except whether the person's record is in it. For this observer only D and D' are possible, and before the release they hold some odds for D against D'. Bayes' rule says that seeing the output y multiplies those odds by the ratio of the two densities at y:

$$\frac{\Pr[D \mid y]}{\Pr[D' \mid y]} = \frac{\Pr[D]}{\Pr[D']} \cdot \frac{p(y)}{q(y)}. \tag{3}$$

So the ratio $p(y)/q(y)$ is exactly how far the output moves the observer's belief. Its logarithm is the **privacy loss** at y. The note uses the logarithm because later sections combine many releases, and the logarithm turns a product of their ratios into a sum:

$$L(y) = \ln \frac{p(y)}{q(y)}. \tag{4}$$

A loss of 0 means that the output is equally likely under both datasets and teaches the observer nothing. A loss of 1 multiplies the odds for D by $e \approx 2.7$, and a negative loss moves the odds towards D'. If $q(y) = 0$ while $p(y) > 0$, the output is impossible under D' and proves that the data are D, so we set $L(y) = +\infty$.

In the counting example, take Laplace noise of scale $b = 1$ and true counts $f(D) = 1$ and $f(D') = 0$. The constants $1/(2b)$ cancel in the ratio, and the logarithm of a ratio of exponentials is the difference of the exponents, so $L(y) = |y| - |y - 1|$. Table 3 evaluates it at a few outputs.

Table 3: The privacy loss of the Laplace mechanism with scale 1, when the true counts are 1 and 0.

Output y	-2	0	0.5	1	3
Privacy loss $L(y)$	-1	-1	0	1	1

An output of 1 or more points towards D, an output of 0 or less points towards D', and the output halfway between says nothing. The loss never goes beyond ± 1, however extreme the output, and Section 2 shows that this bound is what makes the mechanism private.

1.3 The privacy loss as a random variable

When the data are D, the output Y is drawn from P, so its privacy loss $L(Y)$ is a random variable. Every definition in this note bounds some feature of its distribution, such as its largest value or its average. Each definition also applies to the swapped pair (D', D), whose loss is $-L$, so it constrains small values of L as well.

A density can change on a set of outputs of probability zero without a change to any probability. So a statement about $L(y)$ can only hold for almost every y, that is, except on a set of outputs with probability zero, and such a set cannot affect any guarantee.

2 Pure differential privacy

What is the strongest guarantee you can ask for, and what does it cost over many releases? The natural answer to the first question is a privacy loss that is small at every output.

2.1 The definition and what it says about the loss

The definition is usually stated for events rather than single outputs. An **event** S is a set of possible outputs, such as "the released count is above 100".

Definition 2.1 (Pure differential privacy)**.** Let $\varepsilon \geq 0$. A mechanism satisfies **ε-differential privacy (DP)**, and is called ε-DP, if for every pair of neighbouring datasets $D \sim D'$, in either order, and every event S,

$$\Pr_P[S] \leq e^{\varepsilon} \Pr_Q[S]. \tag{5}$$

In words, no event can be much more likely under one dataset than under its neighbour. For small ε the factor e^{ε} is about $1 + \varepsilon$, so with $\varepsilon = 0.1$ no event becomes more than about 10% more likely. Because the definition covers both orders of the pair, it also gives $\Pr_Q[S] \leq e^{\varepsilon} \Pr_P[S]$.

The definition talks about events, but the observer of Equation 3 updates on one output, through the privacy loss. The next result says that the two views agree, so the rest of the note can work with the loss alone. The probability of an event is the integral of the density over it, so a bound on the density ratio at every output integrates to a bound on every event, and a set of outputs where the ratio is too large is itself an event that breaks the definition.

Theorem 2.1 (Pure DP bounds the loss)**.** *A mechanism is ε-DP if and only if, for every pair of neighbouring datasets, the privacy loss satisfies $-\varepsilon \leq L(y) \leq \varepsilon$ for almost every output y.*

Proof. **From the loss to events.** The upper bound $L \leq \varepsilon$ says that $p(y) \leq e^{\varepsilon} q(y)$ for almost every y. Integrating both sides over any event S gives $\Pr_P[S] \leq e^{\varepsilon} \Pr_Q[S]$, because a set of probability zero contributes nothing to an integral. The lower bound $L \geq -\varepsilon$ says that $q(y) \leq e^{\varepsilon} p(y)$, and its integral gives the other order, $\Pr_Q[S] \leq e^{\varepsilon} \Pr_P[S]$.

From events to the loss. Let B be the set of outputs where $p(y) > e^{\varepsilon} q(y)$. On B the density p is positive, so if B has positive length, it has positive probability under P, and the integrand $p - e^{\varepsilon} q$ is positive throughout it. Its integral over B is then positive:

$$\Pr_P[B] - e^{\varepsilon} \Pr_Q[B] = \int_B \left(p(y) - e^{\varepsilon} q(y)\right) \mathrm{d}y > 0. \tag{6}$$

That contradicts the definition for the event B. So B has probability zero, which means that $L \leq \varepsilon$ for almost every output. The same argument with D and D' swapped gives $L \geq -\varepsilon$. □

2.2 The Laplace mechanism

How much Laplace noise makes a statistic ε-DP? By Theorem 2.1 it is enough to bound the privacy loss at every output. The answer is a noise scale proportional to the sensitivity Δf, the most that one record can change the statistic. The proof first computes the privacy loss, then bounds it.

Theorem 2.2 (Laplace mechanism)**.** *Let f have sensitivity Δf. The mechanism that releases $f(D) + Z$, with noise Z drawn from the Laplace distribution of scale $b = \Delta f / \varepsilon$, is ε-DP.*

Proof. **The log-ratio.** On dataset D the output has the Laplace density centred at $f(D)$, and on D' the same density centred at $f(D')$. Both have the normalizing constant $1/(2b)$, so it cancels in the ratio. What remains is a ratio of two exponentials, whose logarithm is the difference of their exponents:

$$L(y) = \ln \frac{p(y)}{q(y)} = \frac{|y - f(D')| - |y - f(D)|}{b}. \tag{7}$$

The numerator compares how far the output y lies from each of the two true answers.

Bounding it. That difference of distances can never exceed the distance between the two answers. To see this, write $y - f(D') = (y - f(D)) + (f(D) - f(D'))$ and apply the triangle inequality to the sum:

$$|y - f(D')| \le |y - f(D)| + |f(D) - f(D')|. \tag{8}$$

Subtract $|y - f(D)|$ from both sides, and use the definition of the sensitivity:

$$|y - f(D')| - |y - f(D)| \le |f(D) - f(D')| \le \Delta f. \tag{9}$$

Dividing by b and substituting into Equation 7 bounds the loss at every output:

$$L(y) \le \frac{\Delta f}{b} = \varepsilon. \tag{10}$$

Swapping D and D' turns the loss into $-L$, and the same argument shows $-L \le \varepsilon$. Theorem 2.1 turns the two bounds into ε-DP. □

For the counting query, $\Delta f = 1$, so noise of scale $b = 1$ gives 1-DP. Table 3 shows that the bound is reached: every output at or below 0, or at or above 1, has a loss of exactly ± 1.

2.3 Many releases: the losses add

Now release several statistics, one after another. When the analyst may choose each statistic after seeing the earlier answers, the composition is **adaptive**, and the i-th release is a mechanism M_i that depends on the dataset and on the earlier outputs $y_1, \dots, y_{i-1}$, which we call the **history**. The analyst's choices depend only on the history, never directly on the data, so for a given history they are the same on both datasets.

The key fact is that the privacy loss of the whole sequence is the sum of the losses of its rounds. The reason is that the probability of a sequence is the probability of the first output, times that of the second given the first, and so on. Write $p_i(y_i \mid \text{history})$ for the density of round i's output given its history on D, and q_i for the same on D'. Then on each dataset the density of the whole sequence is a product:

$$p(y_1, \dots, y_k) = \prod_{i=1}^{k} p_i(y_i \mid y_1, \dots, y_{i-1}). \tag{11}$$

The logarithm of a ratio of two such products is a sum of logarithms of ratios. So the total privacy loss is the sum of the round losses L_i, each computed for its own history:

$$L = \sum_{i=1}^{k} L_i, \qquad L_i = \ln \frac{p_i(y_i \mid \text{history})}{q_i(y_i \mid \text{history})}. \tag{12}$$

Theorem 2.3 (Basic composition)**.** *Suppose that for every history, the i-th release, viewed as a mechanism on the dataset, is ε_i-DP. Then the sequence of k releases is $(\varepsilon_1 + \cdots + \varepsilon_k)$-DP.*

Proof. For every history, round i is ε_i-DP, so Theorem 2.1 gives $|L_i| \le \varepsilon_i$. By Equation 12 the total loss is then at most $\varepsilon_1 + \cdots + \varepsilon_k$ in absolute value, and Theorem 2.1, applied to the whole sequence, turns that bound into the guarantee. □

So a hundred releases at $\varepsilon = 0.1$ each cost $\varepsilon = 10$. That is a weak guarantee, because by Equation 3 a loss of 10 multiplies an observer's odds by $e^{10} \approx 22{,}000$. The theorem is correct, but it charges every round its largest loss, all in the same direction, while the actual loss of a round is random and usually smaller.

The counting example shows by how much. When the data are D, the output is the count 1 plus Laplace noise of scale 1. With probability $\frac{1}{2}$ it lands at or above 1, where the loss is $+1$, and with probability $\frac{1}{2}e^{-1} = 0.184$ it lands at or below 0, where the loss is -1. The outputs in between have losses strictly between -1 and 1, and they add 0.052 to the average. So the loss averages $0.5 - 0.184 + 0.052 = 0.368$, well below its maximum of 1, and it is negative more than a sixth of the time. Over many rounds these losses partly cancel, and Section 5 turns that into a much better bound.

2.4 Post-processing

Analysts round published numbers, combine them and feed them into models. A guarantee is useful only if such a transformation cannot undo it, and it cannot, as long as it does not look at the data again. Apply a function g to the output. The event "g of the output lands in S" is the event "the output lands in the set that g maps into S". The definition applies to that set, so g of the output is ε-DP. A random g whose randomness is independent of the data is an average of fixed functions, and an average keeps the inequality. The same argument works for the guarantee of the next section.

3 Approximate differential privacy

Pure DP needs a bounded privacy loss, but Gaussian noise, the other standard noise, gives an unbounded one. So what guarantee does Gaussian noise give, and what does its new parameter δ measure?

3.1 The Gaussian mechanism's privacy loss

Let the mechanism release $Y = f(D) + Z$, where the noise Z is normal with mean 0 and standard deviation σ. Write $d = f(D) - f(D')$ for the difference between the two true answers. The two normal densities have the same normalizing constant, so the log-ratio is the difference of the exponents:

$$L(Y) = \frac{(Y - f(D'))^2 - (Y - f(D))^2}{2\sigma^2}. \tag{13}$$

When the data are D, the output is $Y = f(D) + Z$, so $Y - f(D) = Z$ and $Y - f(D') = Z + d$. Substitute these and expand $(Z + d)^2 = Z^2 + 2dZ + d^2$, and the Z^2 terms cancel:

$$L(Y) = \frac{(Z + d)^2 - Z^2}{2\sigma^2} = \frac{d^2}{2\sigma^2} + \frac{d}{\sigma^2} Z. \tag{14}$$

So the loss is a constant plus a multiple of the normal noise Z, which makes it normal too. Its mean is the constant $d^2/(2\sigma^2)$. Its standard deviation is the multiplier $|d|/\sigma^2$ times the standard deviation σ of Z, which is $|d|/\sigma$. For a count with $\sigma = 5$ and $d = 1$, the mean is 0.02 and the standard deviation is 0.2, so typical losses are well inside ± 1.

But a normal variable is unbounded, so no ε bounds this loss at every output, and by Theorem 2.1 the Gaussian mechanism is not ε-DP for any finite ε. Large losses do occur, but only with tiny probability, and a useful definition has to allow for them.

3.2 The definition

Definition 3.1 (Approximate differential privacy)**.** Let $\varepsilon \geq 0$ and $0 \leq \delta \leq 1$. A mechanism is (ε, δ)**-DP** if for every pair of neighbouring datasets $D \sim D'$, in either order, and every event S,

$$\Pr_P[S] \leq e^{\varepsilon} \Pr_Q[S] + \delta. \tag{15}$$

The only change from Definition 2.1 is the additive slack δ. It lets an event be more likely under D than the factor e^{ε} allows, by at most δ in probability.

The slack must be far below one over the number of people in the data. To see why, take a mechanism that publishes each person's record in full, independently with probability δ. Unless the extra person's record is published, the output has the same distribution under D and D', and that record is published with probability δ, so the mechanism is $(0, \delta)$-DP. Yet on a dataset of n people it publishes about $n\delta$ records in full, which is why practitioners set δ to 10^{-5} or smaller.

3.3 What δ measures

Given a mechanism and an ε, what is the smallest δ that works? It is the largest value of $\Pr_P[S] - e^{\varepsilon} \Pr_Q[S]$ over events S, and Section 9 uses it for the exact guarantee of the Gaussian mechanism. The idea is that the best event for breaking the inequality collects every output where p exceeds $e^{\varepsilon} q$, and nothing else. To state the result, write $u_+ = \max(u, 0)$ for the positive part of a number, and $\Pr_P[L > \varepsilon]$ for the probability under P of the outputs where the loss exceeds ε.

Theorem 3.1 (The exact δ)**.** *For a pair of output distributions P and Q, the largest value of $\Pr_P[S] - e^{\varepsilon} \Pr_Q[S]$ over all events S is*

$$\int \big(p(y) - e^{\varepsilon} q(y)\big)_+ \, \mathrm{d}y. \tag{16}$$

It can also be written in terms of the privacy loss in two ways:

$$\Pr_P[L > \varepsilon] - e^{\varepsilon} \Pr_Q[L > \varepsilon] = \mathbb{E}_P\Big[\big(1 - e^{\varepsilon - L}\big)_+\Big]. \tag{17}$$

The mechanism is (ε, δ)-DP exactly when this quantity is at most δ for every pair of neighbouring datasets, in both orders.

Proof. **The best event.** Write the quantity to maximize as an integral over the event:

$$\Pr_P[S] - e^{\varepsilon} \Pr_Q[S] = \int_S (p(y) - e^{\varepsilon} q(y)) \, \mathrm{d}y. \tag{18}$$

A set of outputs where the integrand is negative lowers the integral when it joins S, and a set where it is positive raises it. So the integral is largest for the event where the integrand is positive, which is the event $\{L > \varepsilon\}$, and its value there is the integral of the positive part.

The first form. Split that integral into its two terms. The integral of p over $\{L > \varepsilon\}$ is $\Pr_P[L > \varepsilon]$, and the integral of $e^{\varepsilon} q$ over it is $e^{\varepsilon} \Pr_Q[L > \varepsilon]$.

The second form. On the event $\{L > \varepsilon\}$ the definition of the loss gives $q = e^{-L} p$, so the integrand is $p\,(1 - e^{\varepsilon - L})$. Its integral is the average under P of $1 - e^{\varepsilon - L}$ over the outputs where that factor is positive, which is $\mathbb{E}_P[(1 - e^{\varepsilon - L})_+]$. Outputs where $q = 0$ have $L = +\infty$ and count with factor 1. □

This quantity is the **hockey-stick divergence**, written $E_{e^{\varepsilon}}(P \| Q)$, after the graph of $u \mapsto u_+$. Its second form says that δ is an average over the outputs with a loss above ε, with each output discounted by the factor $1 - e^{\varepsilon - L}$. That factor is close to 0 when the loss only just exceeds ε, and close to 1 when the loss is far above it.

4 Proving a guarantee by showing that large losses are rare

Every guarantee in the rest of the note comes from one argument. You show that the privacy loss exceeds some value ε only with a small probability, and you conclude that the mechanism is (ε, δ)-DP with δ equal to that probability. Why is that conclusion valid, and why does it not work in reverse?

The small probability is not a property of the mechanism but a number you choose when you apply a theorem. The note writes it δ', to keep it apart from the δ of a guarantee that a mechanism already has, and in the final guarantee it takes the place of δ. The proof splits an event into the outputs with a small loss, where the pure guarantee holds, and the rest, which is rare.

Lemma 4.1 (Rare large losses give approximate DP)**.** *Suppose that for every pair of neighbouring datasets, in both orders, the privacy loss satisfies $\Pr_P[L > \varepsilon] \le \delta'$. Then the mechanism is (ε, δ')-DP.*

Proof. Fix an event S. Split it into the outputs with a loss of at most ε and those with a larger loss:

$$S_{\text{small}} = S \cap \{L \le \varepsilon\}, \qquad S_{\text{large}} = S \cap \{L > \varepsilon\}. \tag{19}$$

On S_{small} the density bound $p \le e^{\varepsilon} q$ holds, because that is what $L \le \varepsilon$ means. Outputs where $q = 0$ and $p > 0$ have an infinite loss, so they all lie in S_{large}. The integral of the density bound over S_{small} gives

$$\Pr_P[S_{\text{small}}] \le e^{\varepsilon} \Pr_Q[S_{\text{small}}]. \tag{20}$$

Since S_{small} is part of S, its probability under Q is at most $\Pr_Q[S]$, so $\Pr_P[S_{\text{small}}] \le e^\varepsilon \Pr_Q[S]$. The other part is rare by assumption, with $\Pr_P[S_{\text{large}}] \le \Pr_P[L > \varepsilon] \le \delta'$. Adding the two bounds gives $\Pr_P[S] \le e^\varepsilon \Pr_Q[S] + \delta'$. □

The lemma does not work in reverse, which is why Section 5.7 needs a separate argument. A mechanism can be (ε, δ)-DP while its loss exceeds ε far more often than with probability δ, because of the discount in Theorem 3.1. A tail probability counts every output with a loss above ε in full, while δ counts an output whose loss only just exceeds ε at a tiny fraction of its probability. The mechanism of Table 4, with two outputs and some small δ, makes this concrete.

Table 4: A mechanism that is $(0, \delta)$-DP, although its loss is positive half the time.

Output	Probability under P	Probability under Q
0	$\frac{1}{2}$	$\frac{1}{2} + \delta$
1	$\frac{1}{2}$	$\frac{1}{2} - \delta$

Take $\varepsilon = 0$, so that the integral in Theorem 3.1 becomes a sum over the two outputs. Output 0 adds nothing, because it is less likely under P than under Q. Output 1 adds the difference $\frac{1}{2} - (\frac{1}{2} - \delta) = \delta$. So the pair is $(0, \delta)$-DP in this order, and the same computation gives δ in the other order. Yet output 1, which has probability $\frac{1}{2}$, has a positive loss of $\ln(\frac{1}{2}/(\frac{1}{2} - \delta))$, about 2δ, which Theorem 3.1 discounts almost entirely. So an (ε, δ)-DP guarantee does not say that the loss exceeds ε with probability at most δ, and a proof must never assume it.

5 Advanced composition

Theorem 2.3 charged k releases at ε each a total of $k\varepsilon$, as if every round reached its largest loss and all the losses pushed in the same direction. How large is the total loss really, with high probability?

5.1 The idea

Think of the total loss Equation 12 as a random walk with k steps, each between $-\varepsilon$ and ε. Two facts keep the walk short. First, each step is small on average. The average loss is never negative, because outputs that favour D are more likely when the data are D. But Step 1 below shows that it is at most about $\varepsilon^2/2$, far below ε when ε is small, so the averages of the k steps add up to about $k\varepsilon^2/2$.

Second, each step fluctuates around its average, and the fluctuations partly cancel. A sum of k independent fluctuations, each up to ε in size, typically has size about $\varepsilon\sqrt{k}$, not $k\varepsilon$. The rounds here need not be independent, since each may depend on the earlier outputs, but Step 3 shows that the same bound holds. So the total loss is at most about $k\varepsilon^2/2 + \varepsilon\sqrt{k}$, with the second term times a factor that grows slowly as the chosen slack δ' shrinks.

Theorem 5.1 (Advanced composition)**.** *Suppose that for every history, each of k releases is ε-DP. Then for every δ' between 0 and 1, the sequence of releases is $(\varepsilon_{\text{total}}, \delta')$-DP with*

$$\varepsilon_{\text{total}} = \varepsilon\sqrt{2k\ln(1/\delta')} + \frac{k\varepsilon(e^{\varepsilon}-1)}{2}. \tag{21}$$

The proof has four steps. Step 1 bounds the average loss of one round, and Step 2 states two probability tools. Step 3 shows that the sum of the fluctuations is rarely large, and Step 4 adds the averages back and uses Lemma 4.1.

5.2 Step 1: the average loss of one round is small

Fix one round and its history, and write P and Q for that round's two output distributions. The average privacy loss under P is

$$\mathbb{E}_P[L] = \int p(y)\ln\frac{p(y)}{q(y)}\,\mathrm{d}y. \tag{22}$$

This average is the **Kullback–Leibler divergence** of P from Q, written $D_{\text{KL}}(P\|Q)$. What ε-DP controls is the likelihood ratio $R = p/q$, so the first move writes the average in terms of R. Under ε-DP, q is positive wherever p is, so we may divide by it, and $p = qR$ turns the integral into an average under Q:

$$\mathbb{E}_P[L] = \int q(y)\,R(y)\ln R(y)\,\mathrm{d}y = \mathbb{E}_Q[R\ln R]. \tag{23}$$

Two facts about R are now available. Because the round is ε-DP, Theorem 2.1 puts R between $r_- = e^{-\varepsilon}$ and $r_+ = e^{\varepsilon}$. And its average under Q is one, because $\mathbb{E}_Q[R] = \int q\,(p/q) = \int p = 1$. The question is how large $\mathbb{E}_Q[R\ln R]$ can be under these two constraints.

The function $r \mapsto r\ln r$ is convex, since its second derivative is $1/r > 0$, and a convex function lies below its chord, the straight line that joins two of its points. So for every R between r_- and r_+, the value $R\ln R$ is at most the chord from r_- to r_+, which weights the two endpoints by where R sits between them:

$$R\ln R \le \frac{r_+ - R}{r_+ - r_-}\,r_-\ln r_- + \frac{R - r_-}{r_+ - r_-}\,r_+\ln r_+. \tag{24}$$

The right-hand side is a straight line in R, so its average under Q is the same line at the average of R, which is 1:

$$\mathbb{E}_Q[R\ln R] \le \frac{(r_+ - 1)\,r_-\ln r_- + (1 - r_-)\,r_+\ln r_+}{r_+ - r_-}. \tag{25}$$

It remains to simplify this fraction. In the numerator, substitute $\ln r_- = -\varepsilon$ and $\ln r_+ = \varepsilon$, expand, and use $r_-r_+ = 1$:

$$\varepsilon((1-r_-)\,r_+ - (r_+-1)\,r_-) = \varepsilon(r_+ + r_- - 2\,r_-r_+) = \varepsilon\,(r_+ + r_- - 2). \tag{26}$$

So the bound is $\varepsilon\,(r_+ + r_- - 2)/(r_+ - r_-)$. Both parts factor in terms of $s = e^{\varepsilon/2}$, because $r_+ + r_- - 2 = (s - 1/s)^2$ and $r_+ - r_- = (s - 1/s)(s + 1/s)$, as expanding the right-hand sides shows. Cancel the common factor $s - 1/s$, write s out, and use the definition of the hyperbolic tangent, $\tanh u = (e^u - e^{-u})/(e^u + e^{-u})$:

$$\mathbb{E}_P[L] \le \varepsilon\,\frac{s - 1/s}{s + 1/s} = \varepsilon\,\frac{e^{\varepsilon/2} - e^{-\varepsilon/2}}{e^{\varepsilon/2} + e^{-\varepsilon/2}} = \varepsilon \tanh\frac{\varepsilon}{2}. \tag{27}$$

Because $\tanh u \le u$ for $u \ge 0$, the bound is at most $\varepsilon^2/2$, the size the idea promised. Theorem 5.1 uses a looser form, following the literature. Multiplying the numerator and denominator of the middle fraction by $e^{\varepsilon/2}$ gives $\tanh(\varepsilon/2) = (e^\varepsilon - 1)/(e^\varepsilon + 1)$, and since $e^\varepsilon + 1 \ge 2$, the bound is at most $\varepsilon(e^\varepsilon - 1)/2$. In the counting example of Section 2.3, with $\varepsilon = 1$, the exact average loss is 0.368, the bound Equation 27 gives 0.462, and the looser form gives 0.859.

5.3 Step 2: two probability tools

Step 3 has to show that a sum of random fluctuations is rarely large. The standard tool is the moment generating function $\mathbb{E}[e^{\lambda X}]$ of a random variable X, as a function of a number λ. It suits sums because the exponential of a sum is a product of exponentials, which Step 3 averages one factor at a time.

The first turns a bound on the moment generating function into a bound on a tail probability. For every $\lambda > 0$, the event $X \ge t$ is the same as the event $e^{\lambda X} \ge e^{\lambda t}$. Markov's inequality, applied to the positive variable $e^{\lambda X}$, bounds the probability of that event by $\mathbb{E}[e^{\lambda X}]/e^{\lambda t}$. This is **Chernoff's bound**:

$$\Pr[X \ge t] \le e^{-\lambda t}\,\mathbb{E}[e^{\lambda X}] \qquad \text{for every } \lambda > 0. \tag{28}$$

The second, **Hoeffding's lemma**, bounds the moment generating function of a bounded variable with mean zero, and this note cites it without proof (Boucheron, Lugosi and Massart, 2013, Lemma 2.2). If X has mean 0 and always lies between two numbers ℓ and h, then for every real λ

$$\mathbb{E}[e^{\lambda X}] \le \exp\left(\frac{\lambda^2 (h - \ell)^2}{8}\right). \tag{29}$$

A normal variable with mean zero and standard deviation $(h - \ell)/2$ has exactly this moment generating function. So the lemma says that a bounded variable has tails no heavier than that normal variable. The lemma is about a single distribution, so it also applies to the distribution given a history, as long as the mean given the history is zero and the history fixes ℓ and h.

5.4 Step 3: the fluctuations rarely add up to much

All probabilities in this step are for the data D, so the outputs are drawn from P. Let m_i be the average loss of round i given its history. Step 1 bounds it, in the form that Theorem 5.1 uses:

$$m_i \le m_0, \qquad m_0 = \frac{\varepsilon(e^{\varepsilon}-1)}{2}. \tag{30}$$

The fluctuation of round i is $X_i = L_i - m_i$, and it has mean zero given the history. The round loss L_i lies between $-\varepsilon$ and ε, and m_i is a number that the history fixes. So the fluctuation lies between $-\varepsilon - m_i$ and $\varepsilon - m_i$, an interval of width 2ε that the history fixes. Hoeffding's lemma Equation 29 with $h - \ell = 2\varepsilon$ then gives, for every history,

$$\mathbb{E}\big[e^{\lambda X_i} \mid \text{history}\big] \le e^{\lambda^2\varepsilon^2/2}. \tag{31}$$

The rounds need not be independent, because each may depend on the outputs before it. So the argument averages over one round at a time, from the last. Let $S_j = X_1 + \cdots + X_j$ be the sum of the first j fluctuations. By the tower rule for conditional expectations, an average can be taken in two stages, first over round j given the outputs before it, then over those outputs. In the first stage S_{j-1} is fixed, so it comes out as a factor:

$$\mathbb{E}\big[e^{\lambda S_j}\big] = \mathbb{E}\Big[e^{\lambda S_{j-1}}\,\mathbb{E}\big[e^{\lambda X_j} \mid y_1, \ldots, y_{j-1}\big]\Big]. \tag{32}$$

The inner average is the one that Hoeffding's lemma just bounded, so

$$\mathbb{E}\big[e^{\lambda S_j}\big] \le e^{\lambda^2\varepsilon^2/2}\,\mathbb{E}\big[e^{\lambda S_{j-1}}\big]. \tag{33}$$

Repeating this for $j = k, k-1, \ldots, 1$ multiplies k such factors:

$$\mathbb{E}\big[e^{\lambda S_k}\big] \le e^{k\lambda^2\varepsilon^2/2}. \tag{34}$$

Chernoff's bound Equation 28 turns this into a tail bound. For every $t > 0$ and every $\lambda > 0$,

$$\Pr[S_k \ge t] \le \exp\left(-\lambda t + \frac{k\lambda^2\varepsilon^2}{2}\right). \tag{35}$$

The bound holds for every positive λ, so we pick the one that makes it smallest. The exponent is a parabola in λ, smallest where its derivative $-t + k\lambda\varepsilon^2$ is zero, at $\lambda = t/(k\varepsilon^2)$. At that value the exponent is $-t^2/(k\varepsilon^2) + t^2/(2k\varepsilon^2) = -t^2/(2k\varepsilon^2)$, so

$$\Pr[S_k \ge t] \le \exp\left(-\frac{t^2}{2k\varepsilon^2}\right). \tag{36}$$

To make this probability equal to δ', take $t = \varepsilon\sqrt{2k\ln(1/\delta')}$, which is the first term of Theorem 5.1.

5.5 Step 4: add the averages back and conclude

The total loss is the sum of the fluctuations plus the sum of the averages, and each average is at most m_0:

$$L = S_k + \sum_{i=1}^{k} m_i \le S_k + k m_0. \tag{37}$$

So the total loss exceeds $t + k m_0$ only if S_k exceeds t, which by Step 3 has probability at most δ'. The value $t + k m_0$ is exactly $\varepsilon_{\text{total}}$, so

$$\Pr_P[L > \varepsilon_{\text{total}}] \le \delta'. \tag{38}$$

The assumptions are the same for the swapped pair (D', D), so the same bound holds in the other order. Lemma 4.1 now gives the $(\varepsilon_{\text{total}}, \delta')$-DP guarantee, which proves Theorem 5.1.

5.6 How much it saves

The first term of Theorem 5.1 is the fluctuation, and the second is the sum of the averages, close to $k\varepsilon^2/2$ when ε is small, because $e^\varepsilon - 1$ is then close to ε. This form of the theorem is due to Dwork and Rothblum (2016). Take $k = 100$ releases at $\varepsilon = 0.1$, and choose $\delta' = 10^{-5}$, so that $\ln(1/\delta') = 11.513$. The first term is then $0.1\sqrt{200 \times 11.513} = 4.80$ and the second only 0.53. Their total of 5.32 is about half of the 10 that basic composition charges.

The saving grows with the number of releases, because the fluctuation grows like $\sqrt{k}$ while basic composition grows like k. But it shrinks as ε grows, because the second term grows faster than ε. At $\varepsilon = 1$, for example, the second term is $k(e-1)/2 = 0.86\,k$, almost the k that basic composition charges in total, and Section 9 shows a case where advanced composition does worse than basic composition for this reason.

5.7 Rounds that already have a δ

The Gaussian mechanism is only (ε, δ)-DP. So to compose Gaussian releases with the theorems above, as Section 9 does, you need versions of them for rounds that each have a slack δ_0. They exist, and they cost an extra $k\delta_0$ in the slack. By basic composition, k rounds that are each (ε, δ_0)-DP are $(k\varepsilon,\, k\delta_0)$-DP. By advanced composition, they are $(\varepsilon_{\text{total}},\, k\delta_0 + \delta')$-DP, with $\varepsilon_{\text{total}}$ as in Theorem 5.1.

The proofs above do not extend directly. Steps 1 and 3 need the loss of every round to lie between $-\varepsilon$ and ε, and Section 4 showed that an (ε, δ_0) guarantee does not even bound how often the loss exceeds ε. So the proofs rest on a decomposition instead, which this note cites (Dwork, Rothblum and Vadhan, 2010, and Kairouz, Oh and Viswanath, 2015). With probability $1 - \delta_0$, the output of an (ε, δ_0)-DP round comes from a pair of distributions that satisfy Definition 2.1, and with probability δ_0 from an arbitrary pair. The arguments above apply to the first part, and the slack pays for the second.

6 Concentrated differential privacy

The proof of advanced composition used only two facts about each round's loss. Its average is small, and its fluctuation around the average has tails no heavier than those of a normal variable. The bound $|L| \le \varepsilon$ served only to derive them. The Gaussian mechanism has both facts directly, because by Equation 14 its loss is normal with average $d^2/(2\sigma^2)$, yet its loss is unbounded. Can the two facts themselves define privacy? Dwork and Rothblum (2016) made them the definition, so that it covers Gaussian noise without a worst case.

6.1 The definition

The second fact needs a name. A random variable X is **τ-subgaussian** if its fluctuation around its mean has a moment generating function no larger than that of a normal variable with mean zero and standard deviation τ:

$$\mathbb{E}\big[e^{\lambda(X-\mathbb{E}[X])}\big] \le e^{\lambda^2\tau^2/2} \quad \text{for every real } \lambda. \tag{39}$$

A normal variable with standard deviation τ meets this with equality, and Hoeffding's lemma Equation 29 says that a variable confined to an interval of width 2ε is ε-subgaussian. The bound for negative λ controls fluctuations below the mean, as the bound for positive λ controls those above it.

Definition 6.1 (Concentrated differential privacy)**.** A mechanism satisfies (μ, τ)**-concentrated differential privacy (CDP)** if for every pair of neighbouring datasets, in either order, the privacy loss L under P has mean at most μ and is τ-subgaussian.

The mean μ bounds the average loss, and the spread τ bounds how far the loss strays from it. The literature also calls this definition mean-concentrated differential privacy.

The Gaussian mechanism satisfies it. Its loss Equation 14 is normal with mean $d^2/(2\sigma^2)$ and standard deviation $|d|/\sigma$, where the difference d between the two true answers is at most the sensitivity Δf in size. So it is $((\Delta f)^2/(2\sigma^2),\ \Delta f/\sigma)$-CDP. Every ε-DP mechanism satisfies it too, because its average loss is at most $\varepsilon\tanh(\varepsilon/2)$ by Equation 27, and its loss lies between $-\varepsilon$ and ε, so Hoeffding's lemma makes it ε-subgaussian. So an ε-DP mechanism is $(\varepsilon\tanh(\varepsilon/2),\ \varepsilon)$-CDP.

6.2 From CDP to approximate DP

A CDP guarantee is useful only if it says how likely large losses are, so that Lemma 4.1 can turn it into an approximate-DP guarantee. Chernoff's bound gives that tail bound, exactly as in Step 3 of Section 5.

Theorem 6.1 (CDP gives approximate DP)**.** *A (μ, τ)-CDP mechanism is, for every δ' between 0 and 1,*

$$\big(\mu + \tau\sqrt{2\ln(1/\delta')},\ \delta'\big)\text{-DP}. \tag{40}$$

Proof. **A tail bound for the fluctuation.** Apply Chernoff's bound Equation 28 to the fluctuation $X = L - \mathbb{E}_P[L]$, and bound its moment generating function by the subgaussian bound. For every $t > 0$ and every $\lambda > 0$,

$$\Pr[L - \mathbb{E}_P[L] \geq t] \leq \exp\left(-\lambda t + \frac{\lambda^2 \tau^2}{2}\right). \tag{41}$$

The exponent is smallest at $\lambda = t/\tau^2$, where it equals $-t^2/\tau^2 + t^2/(2\tau^2) = -t^2/(2\tau^2)$. With $t = \tau\sqrt{2\ln(1/\delta')}$ the bound equals δ'.

From the fluctuation to the loss. The mean $\mathbb{E}_P[L]$ is at most μ, so a loss above $\mu + t$ is a fluctuation above t. Hence $\Pr_P[L > \mu + t] \leq \delta'$, in both orders, and Lemma 4.1 gives the guarantee. □

6.3 Composing CDP mechanisms

What happens when k rounds are each CDP? The argument of Step 3 in Section 5 carries over and gives an approximate-DP guarantee for the whole sequence. It does not give a CDP guarantee for the sequence, and the reason leads to Section 7.

Theorem 6.2 (Composition of CDP)**.** *Suppose that for every history, round i is (μ_i, τ_i)-CDP. Then for every $\lambda > 0$ the total loss satisfies*

$$\mathbb{E}\Big[e^{\lambda(L - \sum_i \mu_i)}\Big] \leq \exp\Big(\frac{\lambda^2}{2}\sum_i \tau_i^2\Big). \tag{42}$$

Consequently, for every δ' between 0 and 1, the sequence is $(\sum_i \mu_i + \sqrt{\sum_i \tau_i^2}\,\sqrt{2\ln(1/\delta')},\ \delta')$-DP.

Proof. **One round.** Let m_i be the mean of round i's loss given its history, which is at most μ_i. For $\lambda > 0$, a swap of m_i for the larger number μ_i lowers the exponent, so it can only make the average smaller. The subgaussian bound then gives

$$\mathbb{E}[e^{\lambda(L_i - \mu_i)} \mid \text{history}] \leq \mathbb{E}[e^{\lambda(L_i - m_i)} \mid \text{history}] \leq e^{\lambda^2\tau_i^2/2}. \tag{43}$$

All rounds. Averaging over one round at a time, as in Step 3 of Section 5, multiplies these bounds and gives the first claim.

The guarantee. The proof of Theorem 6.1 used the moment generating function only for $\lambda > 0$, and only after a shift of the loss by a number at least as large as its mean. The first claim provides exactly that, with $\sum_i \mu_i$ in place of μ and $\sqrt{\sum_i \tau_i^2}$ in place of τ. Repeating that proof gives the second claim. □

For k identical rounds the guarantee is $k\mu + \tau\sqrt{2k\ln(1/\delta')}$. That is the sum of the averages plus a fluctuation that grows like $\sqrt{k}$, the same shape as advanced composition.

Is the sequence itself $(\sum_i \mu_i, \sqrt{\sum_i \tau_i^2})$-CDP? The proof does not show it, because Definition 6.1 centres the loss at its own mean and needs the bound for every λ. The proof's bound centres the total loss at $\sum_i \mu_i$, which is only an upper bound on its mean. And for $\lambda < 0$, the swap of m_i for the larger μ_i raises the exponent, so the first inequality of the proof runs the wrong way. Bun

and Steinke (2016) discuss why composition statements for CDP are delicate. What the proof does show is the statement that composes cleanly, a bound for positive λ only, centred at a fixed number rather than at the mean. Zero-concentrated differential privacy makes exactly that its definition.

7 Zero-concentrated differential privacy

Which definition makes the saving of advanced composition easy to compute? Bun and Steinke (2016) bound the moment generating function of the loss itself, for every positive argument and with no centring. They state the bound through Rényi divergences, a standard family of divergences between P and Q, whose known properties then come for free. So this section first shows that a Rényi divergence is the moment generating function of the privacy loss in disguise.

7.1 Rényi divergence and the privacy loss

For a number $\alpha > 1$, called the **order**, the **Rényi divergence** of P from Q is

$$D_\alpha(P\|Q) = \frac{1}{\alpha - 1} \ln \int q(y) \left(\frac{p(y)}{q(y)} \right)^\alpha \mathrm{d}y. \tag{44}$$

It is $+\infty$ if P gives positive probability to outputs that Q makes impossible. The integral is the average under Q of the likelihood ratio raised to the power α, and the next lemma rewrites that average in terms of the privacy loss.

Lemma 7.1 (Rényi divergence is a moment generating function of the loss)**.** *For every $\alpha > 1$,*

$$D_\alpha(P\|Q) = \frac{1}{\alpha - 1} \ln \mathbb{E}_P\big[e^{(\alpha-1)L}\big]. \tag{45}$$

Proof. By the definition of the privacy loss, the ratio inside the integral is $p/q = e^L$, so its α-th power is $e^{\alpha L}$, and the integral is the average of $e^{\alpha L}$ under Q. Split $e^{\alpha L}$ as $e^L \cdot e^{(\alpha-1)L}$. Since $q\, e^L = q\,(p/q) = p$, the first factor turns the average under Q into an average under P:

$$\int q\, e^{\alpha L} = \int q\, e^L\, e^{(\alpha-1)L} = \int p\, e^{(\alpha-1)L} = \mathbb{E}_P\big[e^{(\alpha-1)L}\big]. \tag{46}$$

Taking the logarithm and dividing by $\alpha - 1$ gives the claim. □

So the Rényi divergence of order α is the logarithm of the moment generating function of the loss at $\lambda = \alpha - 1$, divided by λ. As α decreases to 1, D_α tends to the average loss, the Kullback–Leibler divergence of Section 5.2. As α grows, larger losses get more weight, and as $\alpha \to \infty$, D_α tends to the largest loss, which pure DP bounds. This note cites both limits (van Erven and Harremoës, 2014).

7.2 The definition

Definition 7.1 (Zero-concentrated differential privacy)**.** Let $\xi, \rho \geq 0$. A mechanism is (ξ, ρ)-**zCDP** if for every pair of neighbouring datasets, in either order, and every $\alpha > 1$,

$$D_\alpha(P\|Q) \leq \xi + \rho\,\alpha. \tag{47}$$

When $\xi = 0$ it is called ρ-**zCDP**.

The bound is a straight line in α because that is the shape the Gaussian mechanism produces, as Theorem 7.1 below shows. To see what the line says about the loss, substitute Lemma 7.1 and write $\lambda = \alpha - 1$, so that $\alpha = 1 + \lambda$. Multiplying by λ gives, for every $\lambda > 0$,

$$\ln \mathbb{E}_P\big[e^{\lambda L}\big] \leq \lambda(\xi + \rho(1+\lambda)) = \lambda(\xi + \rho) + \rho\lambda^2. \tag{48}$$

Move the linear term to the left-hand side and take exponentials:

$$\mathbb{E}_P\big[e^{\lambda(L - \xi - \rho)}\big] \leq e^{\rho\lambda^2} \quad \text{for every } \lambda > 0. \tag{49}$$

This is the form that Section 6 found to compose cleanly, a bound for $\lambda > 0$ only, centred at the fixed number $\xi + \rho$ rather than at the mean. Its right-hand side is the moment generating function $e^{\lambda^2 v/2}$ of a normal variable with variance $v = 2\rho$. So in its upper tail the loss behaves like a normal variable with mean $\xi + \rho$ and variance 2ρ.

7.3 The Gaussian mechanism

The Gaussian mechanism adds normal noise of standard deviation σ to a statistic f of sensitivity Δf. For it the bound in Definition 7.1 holds with equality, which makes ρ a natural measure of its privacy.

Theorem 7.1 (Gaussian mechanism)**.** *Releasing $f(D) + Z$, with Z normal of mean 0 and standard deviation σ, is ρ-zCDP with $\rho = (\Delta f)^2/(2\sigma^2)$, and no smaller ρ works.*

Proof. **The moment generating function.** By Equation 14 the loss is normal with mean $d^2/(2\sigma^2)$ and variance d^2/σ^2. A normal variable with mean m and variance v has a logarithmic moment generating function of $\lambda m + \lambda^2 v/2$, so here

$$\ln \mathbb{E}_P\big[e^{\lambda L}\big] = \frac{\lambda d^2}{2\sigma^2} + \frac{\lambda^2 d^2}{2\sigma^2} = \lambda(1+\lambda)\,\frac{d^2}{2\sigma^2}. \tag{50}$$

The divergence. Dividing by $\lambda = \alpha - 1$, and using $1 + \lambda = \alpha$, Lemma 7.1 gives

$$D_\alpha(P\|Q) = \alpha\,\frac{d^2}{2\sigma^2} \leq \alpha\,\frac{(\Delta f)^2}{2\sigma^2}. \tag{51}$$

Both orders, and tightness. Swapping D and D' only changes the sign of d, and the result depends on d^2, so the same bound holds in the other order. Equality holds for a pair with $|d| = \Delta f$, so no smaller ρ works. □

For a count with $\sigma = 5$, each release is 0.02-zCDP, and doubling σ divides ρ by four.

7.4 Composition: the parameters add

Theorem 7.2 (Composition of zCDP)**.** *Suppose that for every history, round i is (ξ_i, ρ_i)-zCDP. Then the sequence of rounds is $(\sum_i \xi_i,\ \sum_i \rho_i)$-zCDP.*

Proof. **One round.** Fix $\alpha > 1$ and write $\lambda = \alpha - 1$. For every history, the definition bounds the divergence of round i by $\xi_i + \rho_i \alpha$. Multiply that bound by λ, take exponentials, and use Lemma 7.1:

$$\mathbb{E}[e^{\lambda L_i} \mid \text{history}] \le e^{\lambda(\xi_i + \rho_i \alpha)}. \tag{52}$$

All rounds. The total loss is the sum of the round losses, by Equation 12. Averaging over one round at a time, as in Step 3 of Section 5, multiplies these bounds:

$$\mathbb{E}_P[e^{\lambda L}] \le e^{\lambda \sum_i (\xi_i + \rho_i \alpha)}. \tag{53}$$

Back to divergences. Taking the logarithm, dividing by λ and applying Lemma 7.1 to the whole sequence gives $D_\alpha \le \sum_i \xi_i + \alpha \sum_i \rho_i$. This holds for every $\alpha > 1$, and the same argument works in the other order, which is the claim. □

No step needed the average of a round given its history, so the obstacle of Section 6 does not arise, and composition is just addition.

7.5 From zCDP to approximate DP

Theorem 7.3 (zCDP gives approximate DP)**.** *A (ξ, ρ)-zCDP mechanism with $\rho > 0$ is, for every δ' between 0 and 1,*

$$(\xi + \rho + 2\sqrt{\rho \ln(1/\delta')},\ \delta')\text{-DP}. \tag{54}$$

Proof. **A tail bound.** Apply Chernoff's bound Equation 28 to $X = L - \xi - \rho$, using Equation 49. For every $t > 0$ and every $\lambda > 0$,

$$\Pr_P[L > \xi + \rho + t] \le \exp(-\lambda t + \rho \lambda^2). \tag{55}$$

The best λ. The exponent is smallest where its derivative $-t + 2\rho\lambda$ is zero, at $\lambda = t/(2\rho)$, which is positive because $\rho > 0$. Its value there is $-t^2/(2\rho) + t^2/(4\rho) = -t^2/(4\rho)$.

The guarantee. With $t = 2\sqrt{\rho \ln(1/\delta')}$, the ratio $t^2/(4\rho)$ equals $\ln(1/\delta')$, so the tail probability is at most δ'. This holds in both orders, and Lemma 4.1 gives the guarantee. □

Composition (Theorem 7.2, with every $\xi_i = 0$) followed by this conversion gives the main formula of the note. A sequence of k rounds, each ρ-zCDP, is

$$(k\rho + 2\sqrt{k\rho \ln(1/\delta')},\ \delta')\text{-DP}. \tag{56}$$

The parameter ρ adds up linearly, and the square root appears only once, in the final conversion. That conversion is not exact, because it loses in two places. Chernoff's bound overestimates the

tail even at the best λ, and Lemma 4.1 counts every output with a loss above ε in full, without the discount of Theorem 3.1. Sharper conversions exist (Bun and Steinke, 2016; Canonne, Kamath and Steinke, 2020).

7.6 Pure DP and CDP give zCDP

Both earlier definitions fit inside zCDP, so you can account for a mix of Laplace and Gaussian releases in one currency and convert once at the end.

Theorem 7.4 (Pure DP gives zCDP)**.** *An ε-DP mechanism is $(\varepsilon^2/2)$-zCDP.*

Proof. **The fluctuation.** The loss lies between $-\varepsilon$ and ε, so its fluctuation $L - \mathbb{E}_P[L]$ has mean zero and lies in an interval of width 2ε. Hoeffding's lemma Equation 29 bounds its moment generating function by $e^{\lambda^2\varepsilon^2/2}$. Adding back the mean, for every $\lambda > 0$,

$$\ln \mathbb{E}_P\big[e^{\lambda L}\big] = \lambda\, \mathbb{E}_P[L] + \ln \mathbb{E}_P\big[e^{\lambda(L - \mathbb{E}_P[L])}\big] \le \lambda\, \mathbb{E}_P[L] + \frac{\lambda^2\varepsilon^2}{2}. \tag{57}$$

The mean. By Equation 27 the mean is at most $\varepsilon \tanh(\varepsilon/2)$, which is at most $\varepsilon^2/2$. Since $\lambda > 0$, this upper bound can replace the mean and keep the inequality:

$$\ln \mathbb{E}_P\big[e^{\lambda L}\big] \le \frac{\varepsilon^2}{2}(\lambda + \lambda^2) = \frac{\varepsilon^2}{2}\,\lambda(1+\lambda). \tag{58}$$

The divergence. Dividing by $\lambda = \alpha - 1$, and using $1 + \lambda = \alpha$, Lemma 7.1 gives $D_\alpha(P\|Q) \le \alpha\varepsilon^2/2$. The same holds in the other order. □

The proof needs the sharper mean bound $\varepsilon \tanh(\varepsilon/2)$, because the looser form $\varepsilon(e^\varepsilon - 1)/2$ exceeds $\varepsilon^2/2$ and would give a slightly worse ρ. With this theorem, the example of Section 5 becomes a zCDP calculation. A hundred 0.1-DP releases are each 0.005-zCDP, so 0.5-zCDP in total, and Equation 56 with $\delta' = 10^{-5}$ gives $\varepsilon = 5.30$. That is slightly better than the 5.32 of advanced composition.

Theorem 7.5 (CDP gives zCDP)**.** *A (μ, τ)-CDP mechanism is $(\max\{0, \mu - \tau^2/2\},\ \tau^2/2)$-zCDP.*

Proof. **The moment generating function.** Split the loss into its mean and its fluctuation. For $\lambda > 0$, the mean is at most μ and the fluctuation is τ-subgaussian, so

$$\ln \mathbb{E}_P\big[e^{\lambda L}\big] = \lambda\, \mathbb{E}_P[L] + \ln \mathbb{E}_P\big[e^{\lambda(L - \mathbb{E}_P[L])}\big] \le \lambda\mu + \frac{\lambda^2\tau^2}{2}. \tag{59}$$

The divergence. Dividing by $\lambda = \alpha - 1$, Lemma 7.1 gives

$$D_\alpha(P\|Q) \le \mu + (\alpha - 1)\frac{\tau^2}{2} = \Big(\mu - \frac{\tau^2}{2}\Big) + \alpha\,\frac{\tau^2}{2}. \tag{60}$$

Definition 7.1 needs a constant term of at least zero. If $\mu < \tau^2/2$, the constant here is negative, and a constant of 0 in its place keeps the bound true while it meets that requirement. □

For the Gaussian mechanism with noise of standard deviation σ, $\mu = \tau^2/2 = (\Delta f)^2/(2\sigma^2)$, so this returns exactly Theorem 7.1. But approximate DP does not fit inside zCDP at all, because a mechanism with $\delta > 0$ may produce an output that is impossible under a neighbouring dataset, and then every Rényi divergence is infinite.

7.7 Post-processing

Nothing computed from a zCDP output can be less private, because a Rényi divergence can only shrink when you apply the same transformation to both outputs, a property called the data-processing inequality, which this note cites (van Erven and Harremoës, 2014). Bun and Steinke (2016) show that CDP, as defined in Definition 6.1, lacks this property, which is another reason to prefer zCDP.

8 Comparing the four definitions

How do the four definitions compare? Each controls a different feature of the privacy loss, and Table 5 puts them side by side with the results that support each entry.

Table 5: The four definitions, what each controls, and how each composes.

Definition	What it bounds	How it composes	Conversion to (ε, δ)-DP	Post-processing
ε-DP	the largest loss, $\lvert L\rvert \le \varepsilon$	ε adds (Theorem 2.3); advanced (Theorem 5.1)	is already DP	yes
(ε, δ)-DP	the exact δ of Theorem 3.1	ε and δ add; advanced with $k\delta_0$ (Section 5.7, cited)	is already DP	yes
(μ, τ)-CDP	the mean and the spread of L	a one-sided bound (Theorem 6.2)	Theorem 6.1	no (cited)
(ξ, ρ)-zCDP	every Rényi divergence, $D_\alpha \le \xi + \rho\alpha$	ξ and ρ add (Theorem 7.2)	Theorem 7.3	yes (cited)

Never compare an ε with a ρ as numbers. They measure different features of the loss, and only the conversions relate them. For example, a ρ of 2 means an ε of about 11.6 at $\delta' = 10^{-5}$, as the next section computes.

9 A hundred Gaussian releases

How much does each route give away on one task? A count with sensitivity $\Delta f = 1$ is released $k = 100$ times, with Gaussian noise of standard deviation $\sigma = 5$. The analyst may choose each release after seeing the earlier ones, as long as each has sensitivity at most 1. The aim is the smallest ε for which the hundred releases together are $(\varepsilon, 10^{-5})$-DP. In the conversions that final slack is the δ' you choose, so $\delta' = 10^{-5}$ and $\ln(1/\delta') = 11.513$.

Through zCDP. By Theorem 7.1 each release is ρ-zCDP with $\rho = 1/(2 \times 25) = 0.02$. By Theorem 7.2 the sequence is 2-zCDP. Then Equation 56 gives

$$\varepsilon = 2 + 2\sqrt{2 \times 11.513} = 2 + 9.597 \approx 11.60. \tag{61}$$

Through CDP. Each release is $(0.02, 0.2)$-CDP, so Theorem 6.2 has $\sum_i \mu_i = 2$ and $\sqrt{\sum_i \tau_i^2} = \sqrt{100 \times 0.04} = 2$, and it gives $\varepsilon \approx 11.60$ again. The two routes agree because Gaussian noise has $\tau^2 = 2\rho$ in every round, so the fluctuation terms of the two conversions are equal.

The exact answer. Theorem 3.1 gives the smallest δ for each ε, and for Gaussian noise it has a closed form. Write $\gamma = |d|/\sigma$ for the distance between the two true answers in units of the noise's standard deviation, and W for a standard normal variable.

Under P, Equation 14 gives $L = \gamma^2/2 + (d/\sigma)(Z/\sigma)$, where Z/σ is standard normal. A standard normal variable is symmetric about zero, so its product with d/σ has the same distribution as its product with γ, whatever the sign of d. So under P the loss has the distribution of $\gamma^2/2 + \gamma W$.

Under Q, the output is $Y = f(D') + Z$, so $Y - f(D') = Z$ and $Y - f(D) = Z - d$. Substitute these into the loss and expand, and the Z^2 terms cancel again:

$$L = \frac{Z^2 - (Z-d)^2}{2\sigma^2} = -\frac{d^2}{2\sigma^2} + \frac{d}{\sigma^2}\, Z. \tag{62}$$

By the same symmetry, under Q the loss has the distribution of $-\gamma^2/2 + \gamma W$. The first form of Theorem 3.1 needs the probability that the loss exceeds ε under each distribution. Under P it is the probability that W exceeds $\varepsilon/\gamma - \gamma/2$, and under Q the probability that W exceeds $\varepsilon/\gamma + \gamma/2$. Write Φ for the standard normal distribution function, so that W exceeds a number c with probability $\Phi(-c)$. Then

$$\delta(\varepsilon) = \Phi\Big(\frac{\gamma}{2} - \frac{\varepsilon}{\gamma}\Big) - e^{\varepsilon}\, \Phi\Big(-\frac{\gamma}{2} - \frac{\varepsilon}{\gamma}\Big). \tag{63}$$

Swapping D and D' changes only the sign of d, so the same formula holds in the other order. The function $\delta(\varepsilon)$ is the **privacy profile** of the mechanism, and Equation 63 is the exact profile of the Gaussian mechanism (Balle and Wang, 2018).

When the hundred releases are fixed in advance, the total loss is a sum of 100 independent losses. Under P each is normal with mean 0.02 and variance 0.04, and means and variances of independent variables add, so the sum is normal with mean 2 and variance 4. That is the distribution of $\gamma^2/2 + \gamma W$ with $\gamma = 2$, and the same argument under Q gives $-\gamma^2/2 + \gamma W$. So the hundred releases have exactly the loss of one Gaussian release with $\gamma = 2$. A bisection on

Equation 63 for $\delta = 10^{-5}$ with $\gamma = 2$ gives $\varepsilon = 10.00$. Releases that the analyst chooses adaptively are no worse. That follows from the composition theorem of Gaussian differential privacy, a framework built on this exact profile, which this note cites (Dong, Roth and Su, 2022).

Through per-release (ε, δ) guarantees. One release has $\gamma = 0.2$, and Equation 63 gives its ε_0 for any per-release slack δ_0. Basic composition, from Section 5.7, splits the total slack evenly, $\delta_0 = 10^{-5}/100 = 10^{-7}$. That gives $\varepsilon_0 = 0.932$ per release and 93.2 in total.

Advanced composition splits the total slack as $k\delta_0 + \delta' = 10^{-5}$. A grid search over the split finds the best at $\delta_0 = 8.5 \times 10^{-8}$ and $\delta' = 1.5 \times 10^{-6}$, which gives $\varepsilon_0 = 0.938$ per release and 121.6 in total. Table 6 collects the four answers.

Table 6: A hundred Gaussian releases of a count, with $\sigma = 5$, by each route.

Route	Per release	Total ε at $\delta = 10^{-5}$
Basic composition	$(0.932,\ 10^{-7})$-DP	93.2
Advanced composition	$(0.938,\ 8.5 \times 10^{-8})$-DP, $\delta' = 1.5 \times 10^{-6}$	121.6
zCDP or CDP, then convert	$\rho = 0.02$	11.60
Exact	$\gamma = 0.2$	10.00

Advanced composition does worse than basic composition here, as Section 5 warned. The per-release ε_0 is close to 1, so its second term $k\varepsilon_0(e^{\varepsilon_0} - 1)/2 = 73.0$ is already most of the basic total. The concentrated definitions avoid the problem because they never replace the Gaussian loss by a worst case. They work with its actual mean of 0.02 and spread of 0.2 per release, and come within 16% of the exact answer.

Every number in Table 6 is a guarantee, an upper bound for every pair of neighbouring datasets, not the loss of a particular run, which is random and usually much smaller.

10 What the note proves and what it cites

Every theorem and lemma of the note is proved in the text, except five cited results whose proofs need more than the note builds. They are Hoeffding's lemma (Section 5) and the decomposition for rounds with a slack (Section 5.7). The others are the limits and data-processing inequality of the Rényi divergence, the failure of post-processing for CDP (both in Section 7), and adaptive composition for Gaussian noise (Section 9). The note does not prove that CDP is closed under composition in the full sense of Definition 6.1, and it does not rely on it.

11 Sources

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